Phase 2 Summary: Mobile Authentication Integration

How Registration Works

The registration process uses a multi-step fragment flow within the RegisterActivity to collect user data (Name, Email, Password, and Role).

- Process: The app performs client-side validation for email formatting and password strength. Once validated, a POST request is sent to the /api/auth/register endpoint.
- Backend: The Spring Boot backend receives the data, hashes the password using BCrypt, and persists the record into the PostgreSQL (Supabase) database via Spring Data JPA.

How Login Works

The login system authenticates users by verifying credentials against the stored database records.

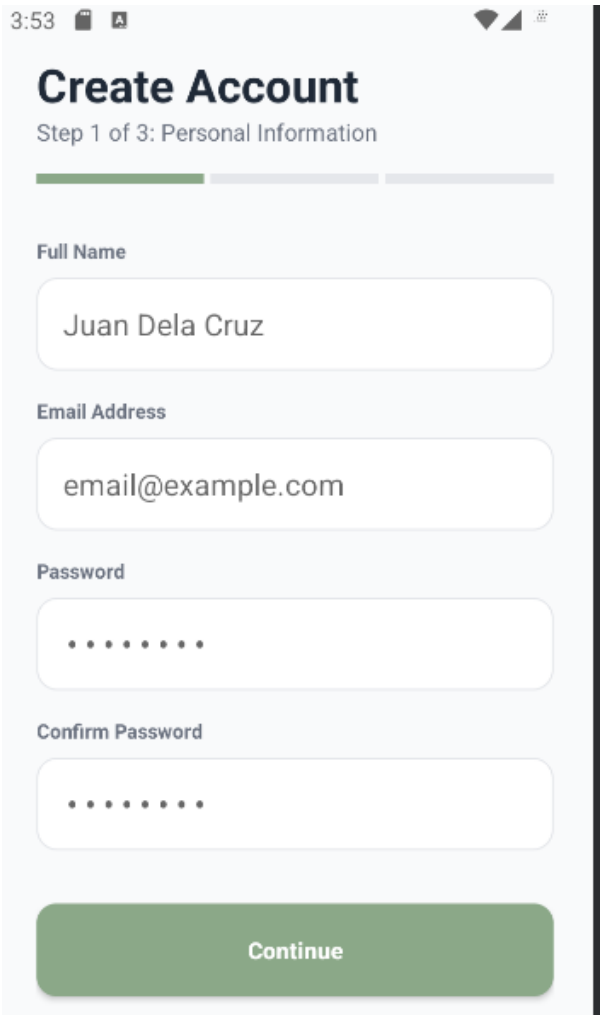
- Process: The LoginActivity sends the user's email and password to the /api/auth/login endpoint.
- Verification: The backend compares the input password with the stored hash. If successful, the server returns the user's profile and role.
- Navigation: The mobile app parses the JSON response and uses role-based logic to redirect the user to either the Patient Dashboard or the Doctor Dashboard.

API Integration Used

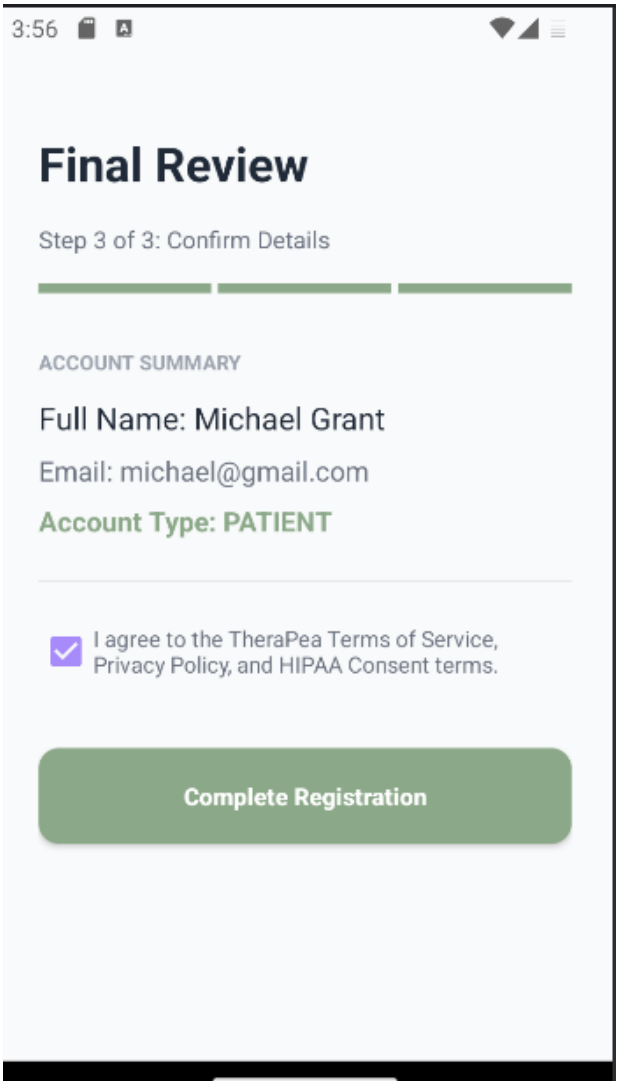
The connection between the Kotlin mobile client and the Spring Boot backend is handled through a RESTful architecture:

- Networking: Retrofit and OkHttp manage asynchronous API calls and network configurations.
- Data Handling: Gson is used for JSON serialization and deserialization, converting server responses into Kotlin data objects.
- Backend Stack: Built with Spring Boot 3.2.5 and Spring Security, the system ensures secure data transmission and high-performance connectivity to the cloud database.

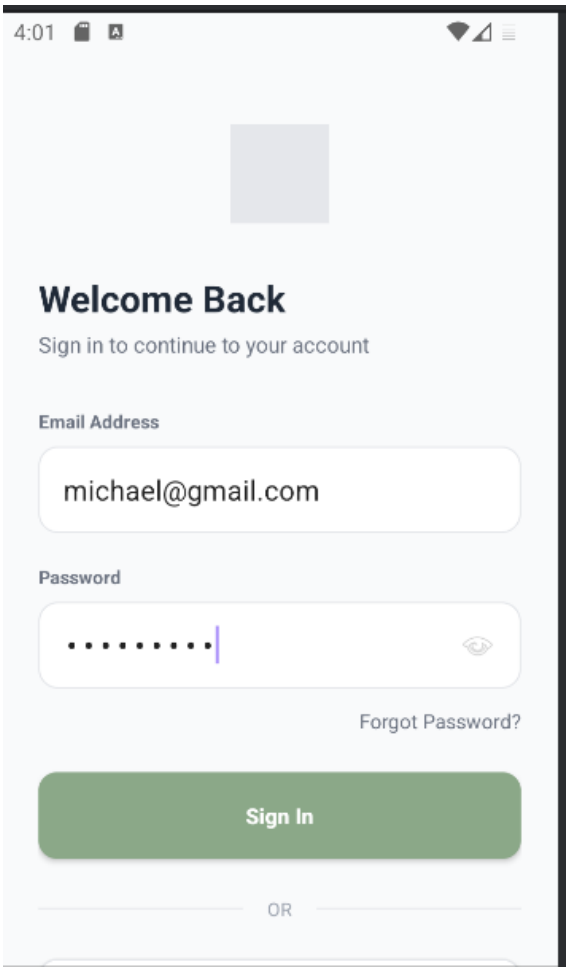
Registration Screen



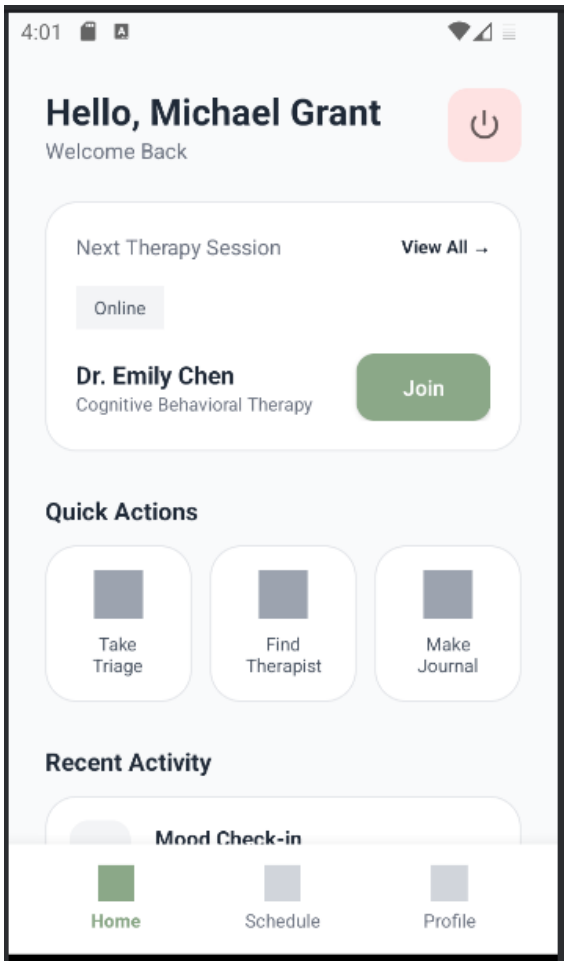
Successful Registration



Login Screen



Successful login



Database Dashboard

email varchar	full_name varchar	password varchar	role varchar	always_logged_in bool
michaelgrantlibato7@gmail.com	Michael Grant Libato	872b2e78-71fc-4bc1-a2cd-74f956f730de	DOCTOR	FALSE
hie@gmail.com	michael grant	$2a$10$AXFzxhUPkdx.9Ti8VE05oeMVw2	DOCTOR	FALSE
ha@gmail.com	hahahh	$2a$10$7nanKNCC6lda8J7H3LEkYul3xH	PATIENT	FALSE
test100@gmail.com	michael	$2a$10$6tRmToT900Ne4/KG815Nfe/1NV	PATIENT	FALSE
michael123@gmail.com	grant	$2a$10$Az/r27SEOc6BLaKlgU1qgOPdrZ6	PATIENT	FALSE
michael@gmail.com	Michael Grant	$2a$10$/yhoBv1mwTpgNTvgx4cR7O4oA	PATIENT	FALSE